Smart Plant Monitoring and Irrigation System

A Project Report
Submitted in partial fulfilment of the
Requirements for the award of the
Degree of

BACHELOR OF SCIENCE (COMPUTER SCIENCE)

Submitted By

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SEAT No. 318

Under the Esteemed Guidance of

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CERTIFICATE

This is to certify that the project entitled, **"Smart Plant Monitoring and Irrigation System"**, is bonafide work of Ms. **Amit Bhual Yadav** bearing Seat No: **318** submitted in partial fulfilment of the requirements for the award of degree of BACHELOR OF SCIENCE in COMPUTER SCIENCE from University of Mumbai.

Internal Guide **Coordinator**

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Date: **College Seal**

DECLARATION

I the undersigned **Amit yadav** hereby declare that the project entitled, **"Smart Plant Monitoring and Irrigation System"** done at **Prahladrai Dalmia Lions College of Commerce & Economics**, has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university.

The project is done in partial fulfilment of the requirements for the award of degree of **BACHELOR OF SCIENCE (COMPUTER SCIENCE)** to be submitted as final semester project as part of our curriculum.

Signature

(Amit Yadav)

Acknowledgement

To list who all have helped me is difficult because they are so numerous and the depths so enormous.
I would like to acknowledge the following as being idealistic channels and fresh dimensions in the completion of this project. I take this opportunity to thank the **University of Mumbai** for giving me the chance to do this project.

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I would also like to express my sincere gratitude towards my project guide **Mr. Anakeet Nena** whose guidance and care made the project successful.

I would like to thank my **College Library**, for having provided various references Books and magazines related to my project.

Lastly, I would like to thank each and every person who directly or indirectly helped me in the completion of the project especially **my Parents and Peers** who supported me throughout my project.

Amit yadav

ROLES AND RESPONSIBILITIES

I **Amit Yadav** take the responsibility of Completing the project pertaining to which we are members. Our Role in this Project involves the development of the whole system from scratch.
All the Roles and Responsibilities will be fulfilled by us within the given time:

Roles and Responsibilities

The project involves designing, assembling, programming and testing a hardware-based plant monitoring and irrigation system.

- Hardware Design: Selecting and connecting the Arduino UNO, soil moisture sensor, relay module and water pump.
- Sensor Integration: Reading soil moisture values and identifying dry soil conditions.
- Controller Programming: Developing the Arduino program to process sensor readings and control irrigation.
- IoT Connectivity: Connecting the controller to an appropriate Wi-Fi/IoT arrangement for remote monitoring.
- Automatic Irrigation: Controlling the water pump according to the soil moisture threshold.
- Testing and Troubleshooting: Checking sensor readings, pump operation, connectivity and overall system performance.
- Documentation: Preparing the project report, circuit diagram, system diagrams and test cases.

ABSTRACT

The Smart Plant Monitoring and Irrigation System is a hardware-based project designed to monitor soil moisture and automate the watering of plants. The system uses an Arduino UNO as its main controller and a soil moisture sensor to measure the moisture level in the soil.

The sensor sends readings to the Arduino UNO, which processes the values and determines whether the soil requires water. When the soil becomes sufficiently dry, the controller activates a water pump through a suitable relay or driver circuit. Once the required moisture condition is reached, the pump is switched off according to the programmed control logic.

The project also supports remote IoT monitoring through a compatible Wi-Fi module or connectivity arrangement. This allows the user to view soil moisture readings and the irrigation status on a supported IoT dashboard when the internet connection is available.

The system aims to reduce unnecessary water usage, minimise manual watering and make plant care more convenient. It can be used for home gardens, potted plants, small nurseries and educational demonstrations.

Overall, this project demonstrates the practical use of sensors, microcontrollers, electronic switching and IoT communication in an automated irrigation system. It focuses on soil moisture monitoring and irrigation without using temperature or air humidity sensors.

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PROJECT PROPOSAL

1.1 Introduction

Plants require the correct amount of water for healthy growth. In many cases, plants are watered manually, which may lead to overwatering or underwatering. Manual watering can also become difficult when the user is busy or away from home.

The Smart Plant Monitoring and Irrigation System is designed to solve this problem using electronic hardware and IoT technology. It measures soil moisture using a sensor and uses an Arduino UNO to control the watering process.

The system can operate automatically according to a predefined soil moisture threshold. It also provides remote monitoring through an IoT platform, allowing the user to check the moisture readings and pump status when connected to the internet.

1.2 Problem Statement

Traditional plant watering depends mainly on manual observation and regular schedules. These methods may not accurately reflect the actual moisture condition of the soil.

Overwatering can waste water and harm plant roots, while insufficient watering can affect plant growth. Users may also find it difficult to monitor plants when they are away from home.

Therefore, there is a need for a simple and affordable system that monitors soil moisture, controls irrigation automatically and provides remote monitoring capabilities.

1.3 Objectives

The main objectives of the project are:

1. To measure soil moisture using a soil moisture sensor.
2. To use Arduino UNO to process the sensor readings.
3. To automate plant watering based on a predefined moisture threshold.
4. To switch the water pump through a suitable relay or driver circuit.
5. To provide remote monitoring through IoT connectivity.
6. To reduce unnecessary watering and manual effort.
7. To develop an affordable prototype for small-scale plant care.

1.4 Scope of the Project

The project focuses on soil moisture monitoring and automatic irrigation for individual plants or a small group of plants.

The scope includes:

- Real-time soil moisture measurement.
- Automatic pump ON/OFF control.
- Remote viewing of sensor readings and pump status.
- Local hardware-based control using Arduino UNO.
- Basic testing of moisture detection and irrigation operation.

The current project does not include temperature measurement, air humidity measurement, automatic fertiliser dispensing or large-scale agricultural control.

1.5 Proposed System

The proposed system consists of a soil moisture sensor connected to Arduino UNO. The Arduino reads the sensor output and compares it with a programmed threshold.

When the soil is dry, the controller activates the pump using a relay or suitable driver. When the soil reaches the programmed moisture condition, the controller switches the pump off.

A compatible Wi-Fi module or connectivity board sends monitoring data to an IoT dashboard. The user can check the readings remotely when connectivity is available.

1.6 Advantages

- Reduces the need for manual watering.
- Helps prevent unnecessary irrigation.
- Provides soil moisture information.
- Supports remote monitoring.
- Uses affordable and commonly available electronic components.
- Demonstrates practical IoT and automation concepts.

SOFTWARE REQUIREMENT SPECIFICATION

2.1 Overall Description

The Smart Plant Monitoring and Irrigation System is an embedded hardware system that uses a sensor, microcontroller and water pump to automate irrigation.

The system receives input from the soil moisture sensor, processes it through Arduino UNO and controls the pump. IoT connectivity is used to send monitoring information to a remote dashboard.

User characteristics

A. User / Plant Owner

- Can monitor soil moisture readings.
- Can check whether the soil is dry or sufficiently moist.
- Can view pump status on the IoT dashboard.
- Can observe the system's irrigation activity.

B. Administrator / Project Maintainer

- Can install and configure the hardware.
- Can adjust the moisture threshold in the program.
- Can test the sensor and pump connections.
- Can troubleshoot the controller and IoT connectivity.

2.2 System Features

A. Soil Moisture Monitoring

The sensor measures the moisture condition of the soil and provides a reading to Arduino UNO.

B. Automatic Irrigation

The controller compares the sensor reading with a programmed threshold and switches the pump on or off according to the control logic.

C. Remote IoT Monitoring

The system sends available sensor readings and pump status to a compatible IoT dashboard using Wi-Fi connectivity.

D. Pump Control

The relay or suitable driver circuit switches the pump according to the controller's output.

E. System Status

The system can report whether the soil requires watering, whether irrigation is active and whether monitoring data is being transmitted.

2.3 Functional Requirements

ID	Requirement
FR1	The system shall read soil moisture sensor values.
FR2	Arduino UNO shall process the sensor readings.
FR3	The system shall compare readings with a configured threshold.
FR4	The pump shall switch on when the programmed dry-soil condition is met.
FR5	The pump shall switch off when the stopping condition is met.
FR6	The system shall transmit monitoring data through compatible IoT connectivity.
FR7	The user shall be able to view readings and pump status remotely.
FR8	The system shall continue local moisture-based control if the IoT connection is temporarily unavailable, provided the controller and hardware remain powered and operational.

2.4 Non-Functional Requirements

- Performance: Sensor readings and pump decisions should be processed within a suitable interval.
- Reliability: The system should operate consistently under normal working conditions.
- Usability: The dashboard should display moisture readings and pump status clearly.
- Maintainability: Sensor thresholds and control logic should be easy to update.
- Safety: The pump and switching circuit must use appropriate voltage and current ratings.
- Availability: Remote monitoring depends on working Wi-Fi, the IoT service and the required power supply.
- Cost-effectiveness: The prototype should use affordable, readily available components.

2.5 External Interface Requirements

Hardware interface

- Arduino UNO
- Soil moisture sensor
- Compatible Wi-Fi module or connectivity board

- Relay module or suitable pump driver
- Water pump
- Appropriate power supplies
- Connecting wires and breadboard or PCB

Software interface

- Arduino IDE
- Arduino-compatible programming libraries
- Compatible IoT platform and dashboard
- Wi-Fi communication software for the selected connectivity hardware

Communication interface

The Wi-Fi arrangement communicates with the IoT service using the protocol supported by the selected platform, such as HTTP or MQTT where available.

2.6 System Constraints

1. Arduino UNO does not provide built-in Wi-Fi connectivity.
2. An additional compatible Wi-Fi module or board is required for remote IoT monitoring.
3. Sensor readings depend on the sensor type, soil condition and calibration.
4. Pump operation depends on the pump's voltage, current and power supply.
5. Remote monitoring requires an available network and supported IoT service.
6. The system monitors soil moisture only; temperature and air humidity are not measured.

2.7 Assumptions

- The soil moisture sensor is installed correctly.
- The controller receives a stable and suitable power supply.
- The water pump and driver circuit are compatible.
- The user has access to a working Wi-Fi connection for remote monitoring.
- The soil moisture threshold is calibrated for the intended plant and soil.

COMPLETE UML SET

Use Case Diagram, ER Diagram, Activity Diagram, Class Diagram, Sequence Diagram and Deployment Diagram. For ahardware project, we can retain the same diagram-based structure but adapt the entities and processes to the actual circuit.

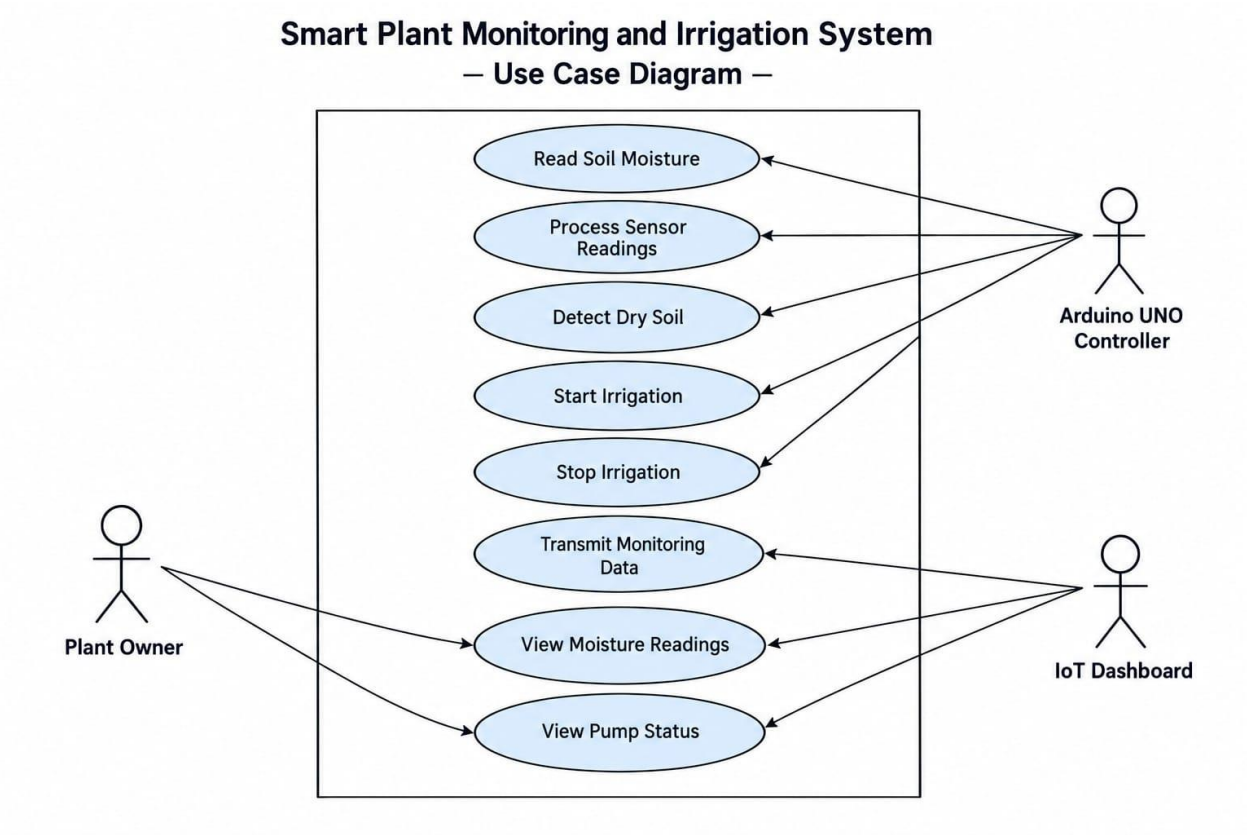
3.1 Use Case Diagram

Actors:

- Plant Owner
- Arduino UNO Controller
- IoT Dashboard

Use cases:

- Read soil moisture
- Process sensor readings
- Detect dry soil
- Start irrigation
- Stop irrigation
- Transmit monitoring data
- View moisture readings
- View pump status



(Use case Diagram)

3.2 ER Diagram / Data Model

An ER diagram can represent the information stored by the IoT dashboard or monitoring application.

Entities and attributes

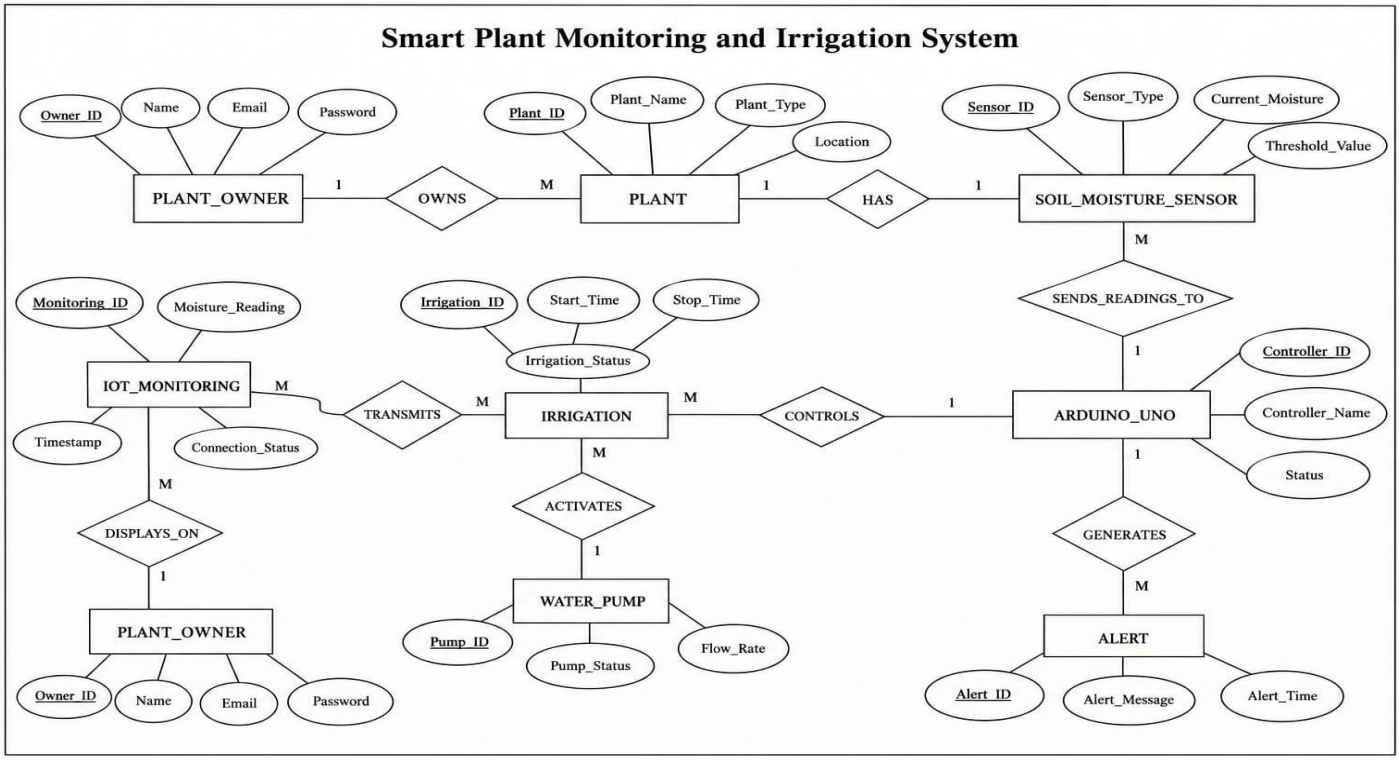


Figure 2: ER Diagram

3.3 Activity Diagram

The activity flow describes the working process.

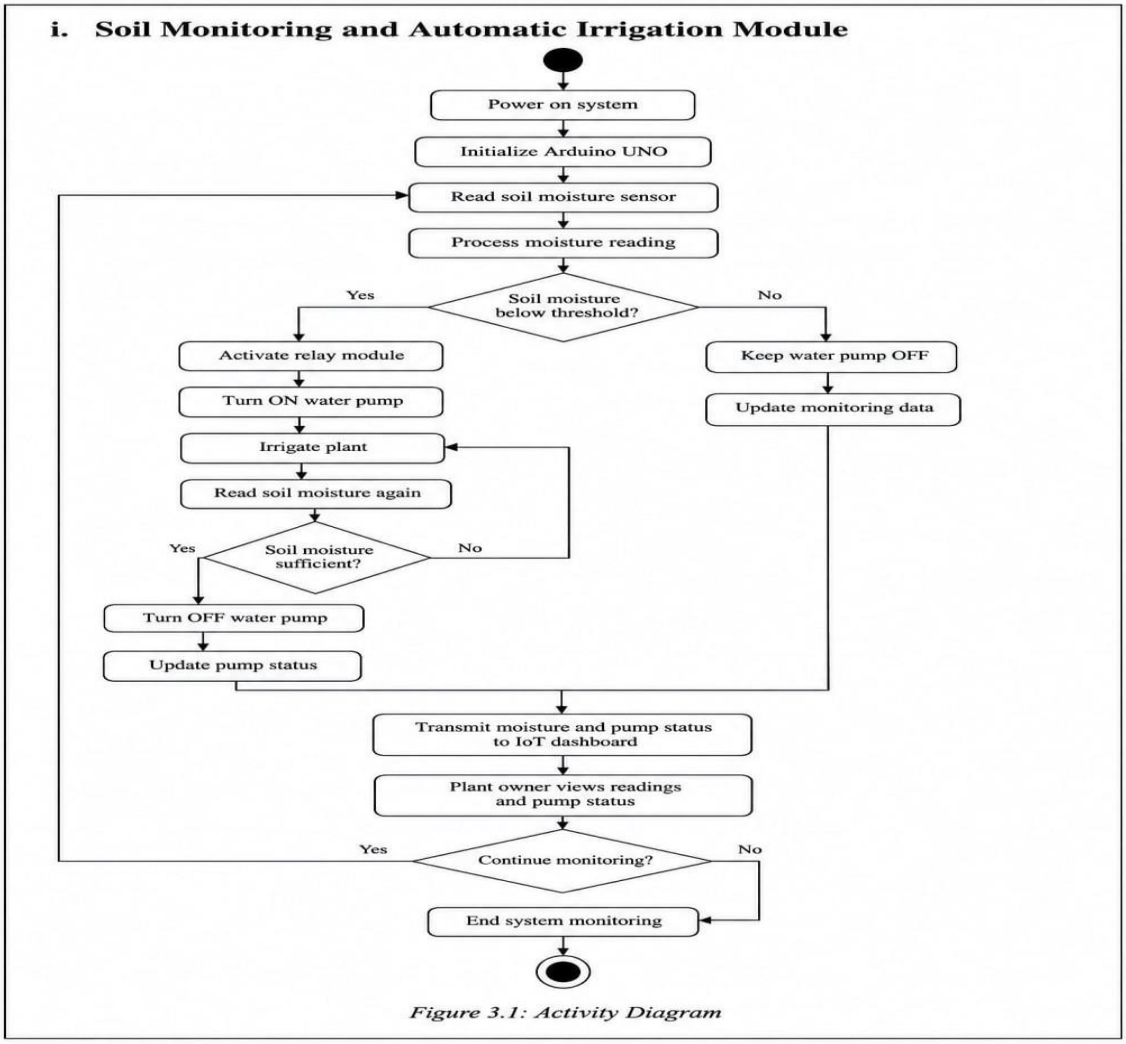
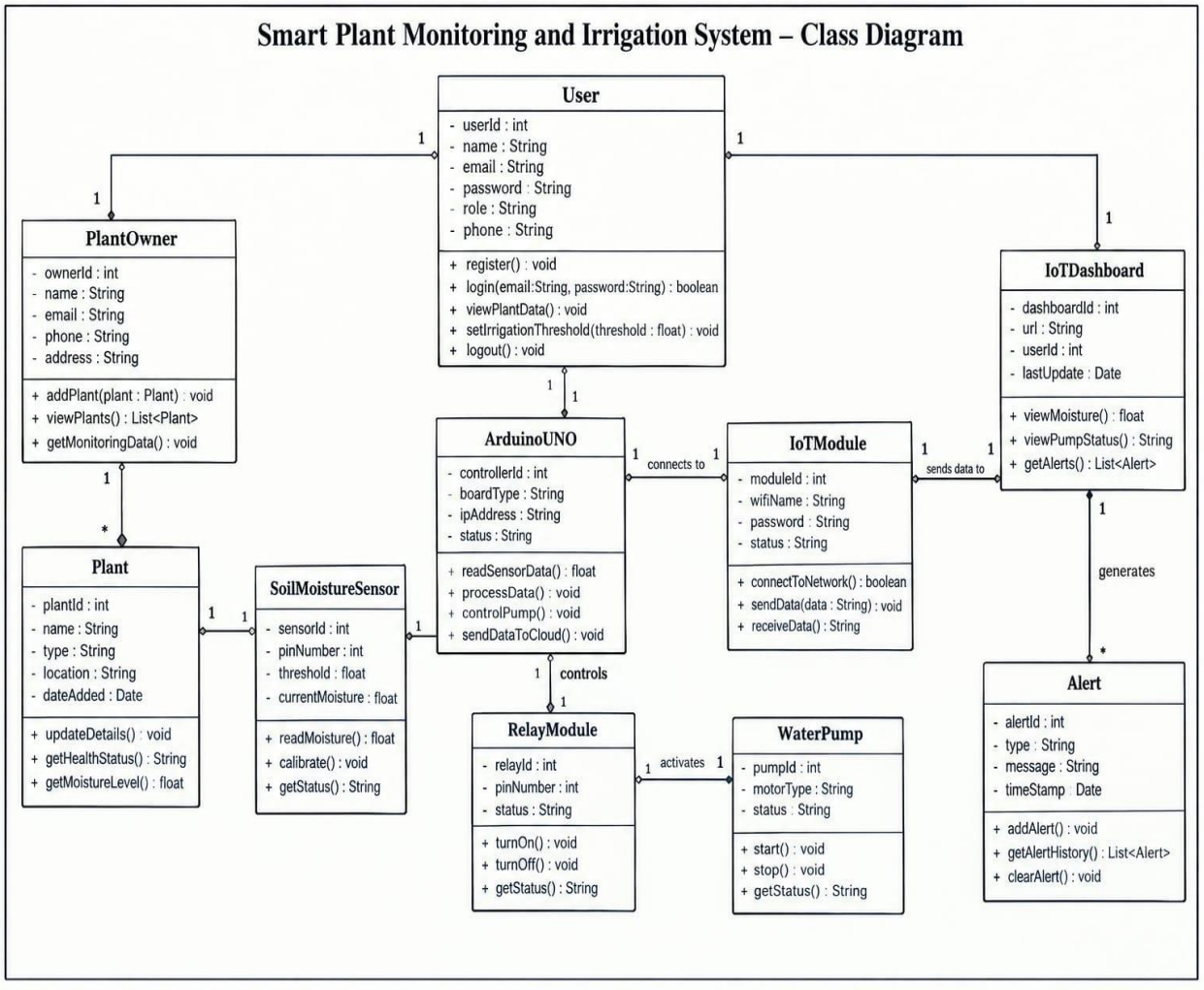


Figure 3.1: Activity Diagram

3.4 Class Diagram

For the software logic running on the controller and the monitoring side, the following conceptual classes can be used.

- MoistureSensor: Reads sensor values.
- PlantController: Processes the reading and applies the irrigation logic.
- PumpController: Switches the pump through the driver circuit.
- IoTMonitor: Sends readings and status to the dashboard.
- SystemSettings: Stores the moisture threshold and other control settings.



ARCHITECTURE DESIGN DOCUMENT

4.1 Hardware Architecture

The hardware architecture explains how the physical

components are connected and interact.

The soil moisture sensor is placed in the soil near the plant roots. Its output is connected to an input of Arduino UNO. The controller processes the readings and generates a control signal for the relay or driver circuit.

The driver circuit operates the water pump using a suitable power source. A Wi-Fi module or compatible board provides the network connection needed to send moisture readings and pump status to the IoT dashboard.

The power arrangement must supply the Arduino, sensor, connectivity module and pump according to their respective electrical requirements.

4.2 Software Architecture

The software architecture consists of the following logical modules:

Sensor Reading Module: Collects soil moisture values.

Processing Module: Converts and evaluates the sensor reading.

Decision Module: Determines whether watering is required.

Pump Control Module: Controls the pump driver.

IoT Communication Module: Transmits readings and status.

Monitoring Dashboard: Displays the latest available data.

The Arduino program repeatedly reads the sensor, makes the irrigation decision and updates the monitoring data.

4.3 Data and IoT Dashboard Design

The monitoring system may display:

Current soil moisture reading.

Dry-soil or sufficient-moisture status.

Water pump ON/OFF status.

Latest update time.

Wi-Fi or device connectivity status.

Historical readings can also be displayed if the selected IoT platform supports data storage and graphs.

4.4 API and Communication Structure

The Wi-Fi connectivity unit transfers information from the controller to the chosen IoT service. Depending on the selected platform, communication may use HTTP requests, MQTT messaging or a platform-specific library.

The message can contain the sensor reading, pump status and device identifier. The dashboard receives this data and presents it in a readable format.

This section should be updated with the actual platform and protocol used in the implemented project.

4.5 Security and Safety Architecture

The IoT monitoring system should use the security features supported by the selected platform, including protected credentials and secure communication where available.

For hardware safety:

Use a relay or driver rated for the pump.

Use a suitable power supply for the pump.

Keep water away from exposed electrical connections.

Protect the controller and circuit from splashes.

Ensure that the pump cannot run dry for extended periods.

Follow proper electrical isolation requirements, especially if mains-powered equipment is involved.

WORKING APPLICATION

Hardware Components and Working System

5.1 Hardware Components Required

Arduino Uno Rev3 - kaufen bei BerryBase

1. Arduino UNO

The main controller of the project. It reads the sensor value, applies the irrigation logic and controls the pump driver.

DIYables Capacitive Soil Moisture Sensor Module for Arduino, ESP32, ESP8266, Raspberry Pi

2. Soil Moisture Sensor

Measures the moisture condition of the soil and sends an electrical signal to the Arduino. Calibration is required to identify dry and moist soil conditions.

Snapklik.com : 4Pcs ESP8266 Serial WiFi Module ESP-01 Updated Wireless Transceiver Board 3.3V For Arduino

3. Wi-Fi Module

Provides network connectivity for remote monitoring. An ESP8266-based module is one possible option, but it must be connected and powered according to its electrical specifications.

Relay Module – MAJESTRONICZ

4. Relay Module or Pump Driver

Allows the Arduino to control the pump without powering it directly from an Arduino output pin. The driver must match the pump's electrical ratings.

Motor bomba de água submersível 3 a 6V DC 120l/h com mangueira de PVC flexível - Robótica Educacional Brasil – Tecnologia, Criatividade e Aprendizagem | Kits STEAM e Cultura Maker

5. Water Pump

Supplies water to the plant when irrigation is required. A low-voltage DC pump is suitable for a small prototype if its power requirements are met.

Breadboard Kit: 400pt + 205 Jumper Wires for Arduino/Raspberry Pi Circuits DIY - Dukes Supplies

6. Supporting Components

Connecting wires, breadboard or PCB, suitable power supply, water tubing and a water container are used to assemble the prototype.

5.2 Hardware Connections

The following is a proposed connection plan. Confirm the exact sensor, Wi-Fi module and relay pinouts before assembling the circuit.

Component	Proposed connection
Soil moisture sensor VCC	Suitable sensor supply
Soil moisture sensor GND	Arduino GND
Soil moisture sensor AO	Arduino A0
Relay/driver control input	Arduino digital output, e.g. D7
Wi-Fi module	Compatible power and communication interface
Water pump	External suitable power supply through relay/driver
Common ground	As required by the selected low-voltage circuit

Important: Do not power the water pump directly from an Arduino pin. Some ESP8266 modules require a regulated 3.3 V supply and are not 5 V tolerant. Their power and logic connections must be designed accordingly. Relay contacts and pump wiring must match the pump's rated voltage and current.

5.3 Working Principle

The system works in the following steps:

The system is powered on.

The soil moisture sensor measures the moisture condition of the soil.

The sensor sends its reading to Arduino UNO.

Arduino compares the reading with the calibrated threshold.

If the soil is dry, the controller activates the pump through the relay or driver.

Water is supplied to the plant.

The controller checks the soil moisture again and stops watering when the programmed stopping condition is reached.

The controller sends the moisture reading and pump status to the IoT dashboard when the network is available.

The process repeats to monitor the plant continuously.

5.4 Algorithm

START

Initialize Arduino

Initialize soil moisture sensor

Initialize pump control

Initialize Wi-Fi/IoT connection

REPEAT:

Read soil moisture sensor

Process the sensor value

IF soil is dry:

Turn pump ON

ELSE:

Turn pump OFF

Update moisture reading

Update pump status

Send data to IoT dashboard if connected

Wait for the next reading

UNTIL system is switched off

STOP

A practical implementation should include sensor calibration, a suitable switching threshold, hysteresis or a dead band to prevent rapid pump switching, and a maximum pump-running time for protection.

5.5 Example Arduino Program

The following is a starter program for local automatic irrigation, not a complete IoT implementation. It assumes an analogue sensor on A0 and a relay module on D7. The moisture threshold and relay polarity must be calibrated for your actual hardware.

```
const int moisturePin = A0;

const int relayPin = 7;


// Example values only: calibrate for your sensor.

const int dryThreshold = 600;


// Many relay modules are active LOW.

// Change these values if your relay behaves differently.

const int PUMP_ON = LOW;

const int PUMP_OFF = HIGH;


void setup() {

Serial.begin(9600);


pinMode(relayPin, OUTPUT);

digitalWrite(relayPin, PUMP_OFF);

}
```

```
void loop() {
int moistureValue = analogRead(moisturePin);

Serial.print("Soil moisture sensor value: ");
Serial.println(moistureValue);

if (moistureValue > dryThreshold) {
digitalWrite(relayPin, PUMP_ON);
Serial.println("Soil is dry: Pump ON");
} else {
digitalWrite(relayPin, PUMP_OFF);
Serial.println("Soil is sufficiently moist: Pump OFF");
}

delay(2000);
}
```

The example assumes that a higher analogue value indicates drier soil, which is common for some sensor arrangements but is not universal. Check the readings from your own sensor in dry and moist soil before selecting the threshold. This program does not yet transmit data to an IoT dashboard.

5.6 IoT Monitoring Implementation

The remote monitoring feature requires an actual Wi-Fi connection and a configured IoT service.

The implementation should:

Connect the Wi-Fi module to the controller.

Configure the required network credentials.

Select an IoT platform and create the relevant device or dashboard.

Send moisture readings and pump status using the platform's supported protocol.

Display the latest reading and status on the dashboard.

Handle connection failures without disabling local irrigation control.

Implementation note: Arduino UNO alone cannot connect directly to Wi-Fi. The specific wiring, program and communication code depend on the Wi-Fi module and IoT platform you choose.

FINAL REPORT

The completed prototype should demonstrate the following results:

Soil moisture readings are collected successfully.

The controller identifies dry-soil conditions using the calibrated threshold.

The pump turns on when watering is required.

The pump turns off when the programmed stopping condition is met.

Moisture readings and pump status appear on the remote dashboard when IoT connectivity is working.

Local irrigation control continues during a temporary network failure, provided the local hardware remains operational.

These are the intended outcomes. Actual results should be recorded after testing the physical prototype.

6.1 Test Cases

Test ID	Test condition	Expected result
TC01	Power on the controller	Arduino starts successfully
TC02	Place sensor in dry soil	Sensor produces a reading corresponding to dry soil
TC03	Place sensor in moist soil	Sensor reading changes appropriately
TC04	Dry-soil threshold is crossed	Pump switches ON
TC05	Moisture reaches stopping condition	Pump switches OFF
TC06	Read sensor through Serial Monitor	Sensor value is displayed
TC07	Wi-Fi connection is available	Device connects to the configured network
TC08	IoT data transmission succeeds	Dashboard updates with current data
TC09	Wi-Fi connection is interrupted	Local irrigation control continues

TC10	Pump driver is tested safely	Driver switches the pump as intended

Record the actual observation and pass/fail result for each test after performing it.

6.2 Limitations

Moisture measurements may vary with soil type and sensor placement.

Sensor calibration is required for reliable irrigation decisions.

Remote monitoring depends on Wi-Fi connectivity and the IoT platform.

The system does not measure temperature or air humidity.

The basic prototype may not detect an empty water tank or blocked pipe unless additional hardware is added.

A small pump-based prototype is not automatically suitable for large farms or multiple irrigation zones.

6.3 Future Scope

The project can be improved in the future by adding:

Water tank level detection.

Multiple soil moisture sensors for multiple plants.

Historical moisture graphs and irrigation logs.

Mobile notifications when the soil becomes dry.

Battery or solar-powered operation.

A manual pump control option on the dashboard.

Improved pump protection and fault detection.

These are possible future additions and are not part of the current system unless implemented.

Conclusion

The Smart Plant Monitoring and Irrigation System demonstrates how hardware and IoT technology can be used to automate plant watering. Arduino UNO processes soil moisture readings and controls a water pump according to a programmed threshold.

The system reduces the need for manual monitoring and supports more controlled use of water. Its IoT component enables remote viewing of sensor readings and pump status when the network is available.

The project provides practical experience in sensor interfacing, microcontroller programming, electronic switching, IoT communication and hardware testing. It can serve as a foundation for more advanced smart gardening and irrigation systems.

Project Repository

If you upload your source code to GitHub, use this section in the report.

GitHub Repository: To be added after uploading the Arduino code, circuit diagram, project documentation and IoT dashboard details.

The repository may contain:

README.md — project introduction, features and setup instructions.

arduino_code/ — Arduino program files.

circuit_diagram/ — circuit and connection diagrams.

iot_dashboard/ — dashboard configuration or related code, if applicable.

test_results/ — photographs and testing observations.

Do not add Wi-Fi passwords, private API keys or other credentials to a public repository.

REFERENCES

Use the documentation relevant to the components and platform actually used in your implementation.

1. Arduino — Arduino Documentation
2. Arduino — Arduino UNO R3 Documentation
3. Espressif Systems — ESP8266 Documentation
4. IoT platform documentation — add the official documentation for your selected platform.
5. Soil moisture sensor datasheet — add the datasheet for your exact sensor model.
6. Relay module or pump driver datasheet — add the documentation for the component used.
7. Water pump datasheet — add the specifications for your actual pump.